



Day and Night — FULL MIND MAP

Topic-wise · Fully Explained · Hinglish · SSC / BPSC / BSSC / Railway

Chapter 02_vi_Day_and_Night

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Equinox — Equal Day-Night

- **Equinox** (equal day-night) saal mein 2 baar hota hai: ~**21 March** aur ~**23 September** — Sun Equator pe vertically chamakta hai. [Q1,Q2]
- Equator pe day-night lagbhag **saal bhar equal (~12-12 hours)** rehta hai — koi distinct winter season nahi hota, sirf wet-dry seasons. [Q3]

Solstices — Longest/Shortest Day

- Summer Solstice (Northern Hemisphere, ~21 June) pe: Equator pe day-length **12 hours** rehta hai (Tropic of Cancer/Capricorn ya Arctic Circle pe nahi). [Q4]
- Northern Hemisphere ka longest day **June ke 2nd half** mein hota hai (~20/21 June). [Q5]
- Northern Hemisphere ka shortest day ~**21/22 December** (Winter Solstice) ko hota hai. [Q6,Q7,Q8]
- Southern Hemisphere ka longest day ~**22 December** ko hota hai (Northern se opposite). [Q9]
- 21 June ko dopahar mein shadow sabse **shortest** hoti hai (Sun Tropic of Cancer pe vertically overhead hota hai). [Q10]
- Summer Solstice har saal **21 June** ko hota hai. [Q11]

21 June — Arctic Circle aur Latitude Sunlight

- 21 June ko Sun **Arctic Circle** pe horizon ke neeche set nahi hota (Midnight Sun phenomenon). [Q12]
- 21 June ko 12 hours se zyada sunlight paane wale latitudes: **Tropic of Cancer aur Arctic Circle** (dono Northern Hemisphere mein, jo Sun ki taraf tilted hoti hai) — Equator aur Tropic of Capricorn **NAHI** (unhe 12 hrs ya kam milta hai). [Q13]



EXAM TRAP

- 21 June ko Equator pe bhi lagbhag exactly 12 hours hi milta hai, **12 se zyada NAHI** — is confusion se bacho. [Q13]

Earth's Rotation Axis — Recent Changes

- Statement analysis: Earth ke rotation/axis mein changes ho rahe hain — **TRUE**; Solar flares/CMEs upper atmosphere ko energy dete hain — **TRUE** (fact-wise correct) lekin yeh axis-change ka primary explanation NAHI hai; Polar ice melting se water Equator ki taraf shift hota hai — **TRUE** AUR yahi axis-change ka scientific explanation hai. [Q14]

 EXAM TRAP

- Do statements factually correct ho sakte hain lekin sirf EK unme se main statement ko explain karta hai — CMEs sirf upper atmosphere tak limited hain, jabki ice-melting se mass-redistribution actually axis ko affect karta hai. [Q14]